

6. Abstraction

 this is mostly theory

What

1. Delivering only **essential information** to the outer world while **masking** the background details.
2. It is a design and programming method that **separates** the *interface* from the *implementation*.
3. Real life e.g.,
 1. various functionalities of AirPods but don't know the actual implementation/working
4. To drive a car, one only needs to know the driving process and not the mechanics of the car engine

How

1. Function implementation is implemented in header files
2. We use the same program without knowing its inside working
3. Eg.
 1. you use `sort()` from `<algorithm>` but you don't know what technique at its backed is used to get you the desirable result

Implementation

1. Grouping data members and member functions into classes using access specifiers.
2. A class can choose which data members are visible to the outside world and which are hidden.